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Submitter Information

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Organization: Project Management Institute (PMI)

General Comment

The Project Management Institute (PMI) hereby submits the following comments to the Office of Science and Technology Policy (OSTP) regarding American leadership in artificial intelligence.

Attachments

AI Executive Order (FINAL)

Michael Kratsios
Office of Science and Technology Policy
National Science Foundation
1600 Pennsylvania Avenue NW
Washington, DC 20500

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

The Project Management Institute (PMI) hereby submits the following comments to the Office of Science and Technology Policy (OSTP) regarding American leadership in artificial intelligence through private sector initiatives.

Background

PMI, founded in the United States in 1969 and proudly headquartered in Newtown Square, Pennsylvania, is the world's leading organization for project management, with over 742,000 members. PMI's Cognitive Project Management for AI (CPMAI) methodology and certification, obtained through its acquisition of AI thought leaders Cognilytica, is a vendor-neutral, data-centric, AI-specific, iterative methodology for running and managing AI, Machine Learning (ML), and cognitive technology projects. PMI is uniquely positioned to contribute to America's continued global dominance in AI through our frameworks, methodologies, and professional certifications.

PMI's Contributions to American AI Leadership

Since 2017, PMI Cognilytica has established itself as an AI thought leader, championing best practices and trustworthy AI implementation in both the public and private sectors. Following the acquisition, PMI continues to uphold and expand this thought leadership.

Seven Patterns of AI Framework

While many applications fall generally under the umbrella of "artificial intelligence," these AI solutions can often look different in practice. To address these differences, PMI Cognilytica identified that AI projects fit within seven main "patterns" of AI. Each pattern comes with its own data requirements, risks, and considerations. By mapping an AI project to one or more of these seven patterns, implementers can accelerate AI technology decisions, more accurately gauge required data, and better scope the project to address specific AI project needs and risks.

This "Seven Patterns of AI" framework identifies the fundamental patterns of AI applications across all AI applications regardless of industry or use case. These patterns include:

1. Hyperpersonalization: Using machine learning to develop unique profiles of individuals that learn and adapt over time, treating each person as an individual rather than as part of a broad category.
2. Conversational/Human Interaction: Enabling machines to communicate with humans through various interfaces in the language of humans.
3. Pattern & Anomaly Detection: Identifying patterns in data and determining whether new data fits existing patterns or represents anomalies.
4. Predictive Analytics & Decision Support: Using historical data to help humans make predictions about future events and support decision-making processes.
5. Recognition: Identify and understand things within otherwise unstructured content such as objects, video, audio, or text.
6. Autonomous Systems: Creating systems that can operate independently with minimal human intervention.
7. Goal-Driven Systems: AI systems that are designed to achieve specific goals by optimizing their actions and find the optimal solution to a problem.

In 2022, PMI's American-made "Seven Patterns of AI" were adopted by the Organization for Economic Co-operation and Development (OECD), an international organization that works with over 100 countries to develop evidence-based policies, as a core component to their definition of AI.¹

Trustworthy AI Framework

PMI's Trustworthy AI Framework provides an approach to reliable and trustworthy AI that is aligned with core American values. The framework addresses five key layers:

1. Ethical AI: Ensuring AI systems participate in society positively, provide human benefit, avoid harm, and maintain human control and agency.
2. Responsible AI: Addressing potential misuse or abuse of AI, including aspects of AI safety, privacy, trust, and human accountability.
3. Transparent AI: Increasing trust of AI systems by providing visibility into system behavior, data configuration, user consent, bias mitigation, and disclosure and consent.
4. Governed AI: Implementing processes, controls, and safeguards on AI systems, including auditability, monitoring, and complying with regulations as well as provide ways to contest or appeal AI decisions for human review.
5. Explainable AI: Reducing the "black box" nature of AI systems by providing means to understand how machines arrive at conclusions and to understand the root cause of any failures.

This framework enables organizations to build AI systems that concurrently deliver business value that is aligned with American values using a flexible, adaptable approach that imposes the least burden on AI application developers.

¹ OECD (2022), "OECD Framework for the Classification of AI systems", OECD Digital Economy Papers, No. 323, OECD Publishing, Paris, <https://doi.org/10.1787/cb6d9eca-en>.

CPMAI Methodology and Professional Certification

The CPMAI methodology provides a structured approach to running and managing AI projects. This vendor-neutral, data-centric, AI-specific, iterative methodology has become the industry standard for successful AI implementation.² By combining best practices from project management with the specific requirements of AI projects, CPMAI addresses the unique challenges of AI development and deployment.

CPMAI was designed to close AI implementation gaps and reduce the high rate of AI project failures. The CPMAI methodology extends familiar, proven approaches such as agile and data-focused frameworks with AI-specific best practices. Through its six-phase approach for AI projects, the CPMAI process focuses on ensuring AI projects have alignment with business objectives and ROI, proper data preparation and governance, robust model evaluation and operationalization, a focus on real-world AI implementation and not proof-of-concept experimentation, and iterative development to adapt to changing data and needs.

Promoting American Values Through Private Sector Leadership

PMI firmly believes that private sector leadership, rather than excessive government regulation, is the most effective way to ensure America's continued preeminence in AI. Our frameworks and methodologies embody core American values, like:

- **Innovation and Entrepreneurship:** Our approaches empower organizations to innovate, creating new opportunities for growth and competitiveness.
- **American Leadership:** We promote American values in AI development, ensuring that American AI systems reflect our nation's values.
- **Market-Driven Solutions:** Our frameworks are developed based on real-world needs and are continuously refined by the market, ensuring their relevance and effectiveness.
- **Professional Excellence:** We elevate the profession of project management and AI development through rigorous standards, professional certifications, and lifelong learning.

Recommendations:

- **Encourage Adoption of Private Sector, Industry-Led Frameworks:** Promote the voluntary adoption of frameworks like the Seven Patterns of AI, Trustworthy AI, and CPMAI methodology, including training and professional certifications as American-led standards for AI.
- **Support Professional Development:** Invest in programs that build America's AI workforce through professional certifications that promote career readiness in AI.

² PMI.org/AI

- **Promote American AI Standards Globally:** Support efforts to establish American-led private sector standards as the global preeminent methodologies for AI.

Conclusion

PMI's work is supporting America's continued preeminence in artificial intelligence through our voluntary frameworks, methodologies, and professional certifications. By promoting private sector-led initiatives that embody American values of innovation and excellence, the US federal government can be assured that the United States remains at the forefront of AI. PMI appreciates the opportunity to provide these comments and would welcome further discussion on how PMI contributes to American leadership in AI. Please feel free to reach out to Manthan Bhatt at [REDACTED]

Sincerely,

[REDACTED]

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Project Management Institute